

Math 427 homework 6 plus

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Theorem 1. Let f be analytic in an open set containing the closed disc $\overline{D}_R(z_0)$. Then f admits a unique power series expansion at z_0 , say $f(z) = \sum_{n=0}^{\infty} c_n(z - z_0)^n$, which converges on $D_{R'}(z_0)$ for some $R' > R$. Then $f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{|w-z_0|=R} \frac{f(w)}{(w-z_0)^{n+1}} dw$ for $n \in \mathbb{N}$. In particular, $c_n = \frac{f^{(n)}(z_0)}{n!}$. If $|f(z)| \leq M$ on the boundary of $\overline{D}_R(z_0)$, then $|f^{(n)}(z_0)| \leq \frac{n!M}{R^n}$ and $|c_n| \leq \frac{M}{R^n}$ for $n \in \mathbb{N}$.

This theorem says if f is analytic on a closed disk, then $f(z)$ for any interior point z is completely determined by f restricted to the boundary.

Problem A If f has a power series expansion at z_0 with radius of convergence R , and if $r < R$, then there is a constant C such that $|f(z) - f(z_0)| \leq C|z - z_0|$ provided $|z - z_0| \leq r$.

proof: cauchy estimates. Consider $f(z) = \sum_{n=0}^{\infty} c_n(z - z_0)^n$ on $D_R(z_0)$ for $z \in \overline{D}_r(z_0)$. Choose s such that $r < s < R$. Then f is analytic on $K = \overline{D}_s(z_0)$. Because f is continuous, $|f|$ admits a maximum on K (or on ∂K). Let $M = \sup_{z \in K} |f(z)|$. By theorem 1, we get $|c_n| \leq \frac{M}{s^n}$ for all $n \in \mathbb{N}$. It follows that

$$\begin{aligned} |f(z) - f(z_0)| &= \left| \sum_{n=1}^{\infty} c_n(z - z_0)^n \right| \\ &\leq \sum_{n=1}^{\infty} |c_n| |z - z_0|^n \\ &= |z - z_0| \left(\sum_{i=0}^{\infty} |c_{i+1}| |z - z_0|^i \right) \\ &\leq |z - z_0| \left(\sum_{i=0}^{\infty} \frac{M}{s^{i+1}} r^i \right) \\ &= \frac{M}{s} |z - z_0| \sum_{i=0}^{\infty} \left(\frac{r}{s} \right)^i \\ &= \frac{M}{s} \frac{1}{1 - \frac{r}{s}} |z - z_0|. \end{aligned}$$

□

Remark. In the proof, $M = \sup_{z \in K} |f(z)| = \sup_{z \in \partial K} |f(z)|$. See Taylor Corollary 3.5.3.

Problem B If $f(z) = \sum a_n(z - z_0)$, then $|f(z) - \sum_{n=0}^k a_n(z - z_0)^n| \leq D_k |z - z_0|^{k+1}$ where D_k is a constant and $|z - z_0| \leq r < R$.

Proof. We use the same idea as in Problem A. Choose some $r < s < R$ and denote $K = \overline{D}_s(z_0)$. Let $M = \sup_{z \in K} |f(z)|$. Then by theorem 1, we get $|c_n| \leq \frac{M}{s^n}$ for all $n \in \mathbb{N}$. It follows that

$$\begin{aligned}
 |f(z) - \sum_{n=0}^k a_n(z - z_0)^n| &= \left| \sum_{n=k+1}^{\infty} c_n(z - z_0)^n \right| \\
 &\leq \sum_{n=k+1}^{\infty} |c_n| |z - z_0|^n \\
 &= |z - z_0|^{k+1} \left(\sum_{i=0}^{\infty} |c_{i+k+1}| |z - z_0|^i \right) \\
 &\leq |z - z_0|^{k+1} \left(\sum_{i=0}^{\infty} \frac{M}{s^{i+k+1}} r^i \right) \\
 &= \frac{M}{s^{k+1}} |z - z_0|^{k+1} \sum_{i=0}^{\infty} \left(\frac{r}{s} \right)^i \\
 &= \frac{M}{s^{k+1}} \frac{1}{1 - \frac{r}{s}} |z - z_0|^{k+1}.
 \end{aligned}$$

□

proof 2 for problem A and B. Let $k \in \mathbb{N}$ be arbitrary. Note that $g_k = \frac{f(z) - \sum_{n=0}^k a_n(z - z_0)^n}{(z - z_0)^{k+1}} = \sum_{n=k+1}^{\infty} c_n(z - z_0)^n$ is defined by a power series expanded at z_0 with radius of convergence R . Indeed, the coefficients of g_k is a k-tail of that of f . So the lim sup is the same. Thus g_k is analytic on $K = \overline{D}_s(z_0)$ for some $r < s < R$. It follows $|g_k|$ is uniformly continuous on K and hence attains a maximum M . That is, $|\frac{f(z) - \sum_{n=0}^k a_n(z - z_0)^n}{(z - z_0)^{k+1}}| \leq M$. □

Problem C For large k , we can estimate D_k by $1/R^k(R - r)$.

Proof. Note $D_k = \frac{M}{s^{k+1}} \frac{1}{1 - \frac{r}{s}} |z - z_0|^{k+1} = \frac{M}{s^k(s-r)}$ for all $r < s < R$. So take $s \rightarrow R$. □